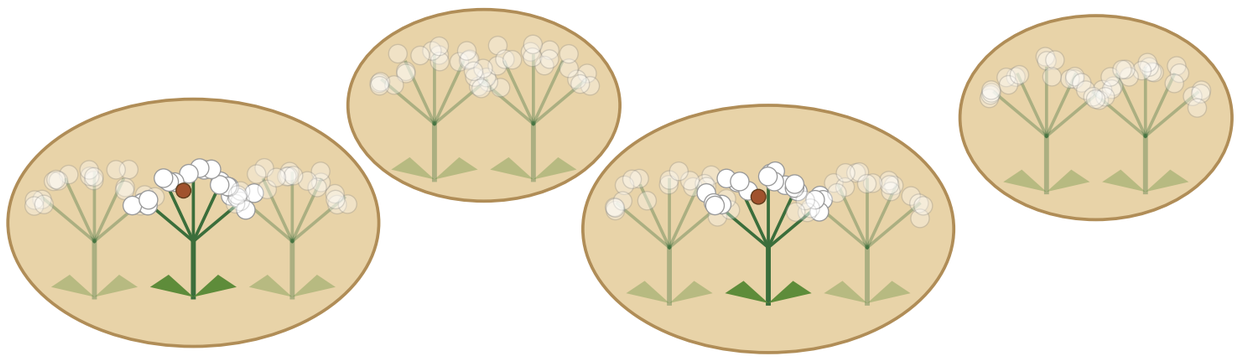
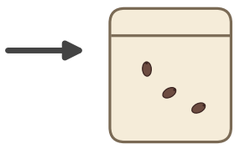
The challenge: very little seed to start with — why?

federally threatened (ESA)

wild populations: limited fertility — few fruits





very little seed

✗ **far too little to restore populations directly**

✗ **collecting more would deplete the populations it comes from**

1. Self-incompatible breeding system: plants reject their own pollen

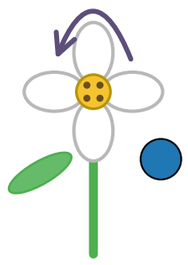


color = S-type
(32 known in LEPA)

under genetic control

self-incompatible

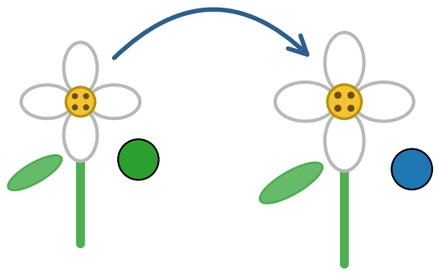
own pollen





no seed

other S-type

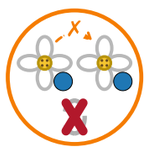




seed

the S-locus (a male + a female gene) gives each plant an S-type: pollen with the flower's own S-type is rejected

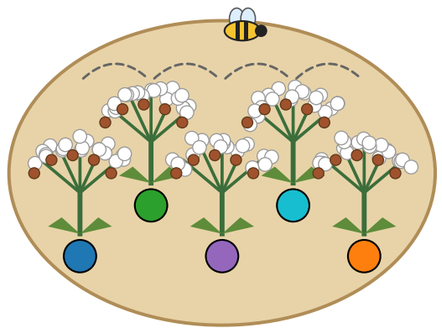
2. Fragmentation → genetic drift → mate limitation → population decline

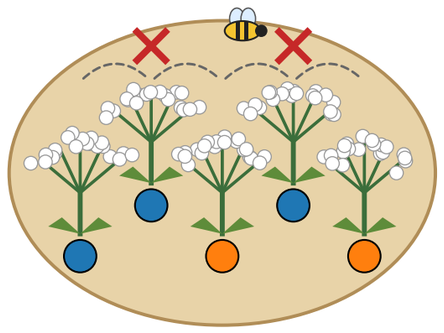


mate limitation

habitat fragmentation → genetic drift

many S-types → **few S-types**









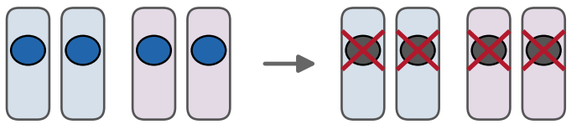
in the field: less seed than plant size predicts

small, isolated populations lose S-types by drift: plants run out of compatible mates and set little seed

3. Rapid adaptation to mate limitation: → increase in seed (short-term gain)

self-compatible

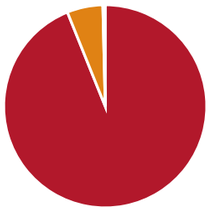
SI **SC**



working S-locus (4 copies)

broken copies

263 plants genotyped



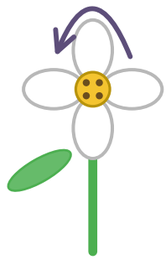
94% self-incompatible

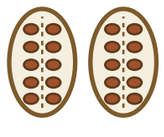
6% partly broken

<1% self-compatible

in the field: more seed than plant size predicts

own pollen





seed ↑

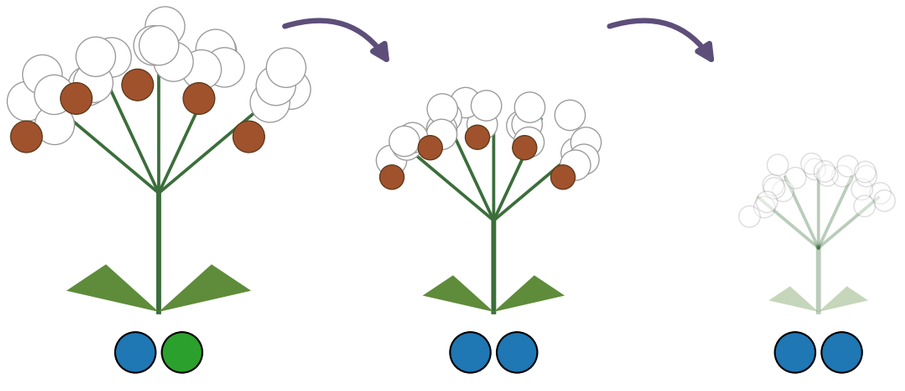
plants with broken S-locus copies accept their own pollen and set seed alone, without mates

4. Long-term cost: inbreeding depression → population decline



inbreeding depression

self *self*



parent

selfed offspring

next generation

seed ↓

survival ↓

genetic diversity ↓
reproduction ↓

selfed offspring grow more alike and weaker, generation after generation